Chapter 4 – Question 23

X_1 = time to completion of road 1. $X_1 \sim Unif(15,20)$. $f(x_1) = 1/5$, $F(x_1) = \frac{x_1 - 15}{5}$

X_2 = time to completion of road 2. $X_2 \sim Unif(15,20)$. $f(x_2) = 1/5$, $F(x_2) = \frac{x_2 - 15}{5}$

Y = time to completion of bridge. $Y \sim Unif(12,19)$. $f(y) = 1/7$, $F(y) = \frac{y - 12}{7}$

a) See course notes for expected value, variance/standard deviation of a uniform random variable.

b) The project is complete once all sub-projects are complete, so the project will take as long as the longest-running sub-project. Time to completion $T = \max(X_1, X_2, Y)$. We wish to find $E(T)$.

Note: $E(T) = \int_{-\infty}^{\infty} t f(t) dt$, so we need to figure out $f(t)$.

We can figure out the cumulative distribution function by taking the product of the corresponding CDFs for each random variable (see lecture notes), evaluated at some time t. Note that we must be extra careful here: the cumulative distribution function will be a piecewise function. In the time range [15,19], all three sub-projects contribute. In the range [19,20], the bridge will have been complete, so only the two roads contribute to the CDF. The project will take at minimum 15 months, and at maximum 20 months.

- $F(t) = F_{X_1}(t) F_{X_2}(t) F_Y(t) = \frac{(t-15)}{5} \frac{(t-15)}{5} \frac{(t-12)}{7} = \frac{1}{175}(t-15)^2 (t-12)$, $15 \le t \le 19$
- $F(t) = F_{X_1}(t) F_{X_2}(t)$, $= \frac{1}{25}(t-15)^2$ $19 \le t \le 20$.

We can figure out the (piecewise) density by differentiating each piece above.

- $f(t) = \frac{1}{175}(t-15)^2 + \frac{2}{175}(t-15)(t-12)$, $15 \le t \le 19$
- $f(t) = \frac{2}{25}(t-15)$, $19 \le t \le 20$

Now, we can calculate $E(T) = \int_{15}^{20} t f(t)$.

- $= \int_{15}^{19} t \times (\frac{1}{175}(t-15)^2 + \frac{2}{175}(t-15)(t-12)) dt$
  $+ \int_{19}^{20} t \times (\frac{2}{25}(t-15)) dt$

After some fairly painful calculation (or by using your trusty TI-89), you should find

- $E(T) = \frac{80}{7} + \frac{527}{75} = \underline{18.45}$ months.

c) Now that we have the CDF, we can calculate $P(T \le 18)$ quite readily.

$P(T \le 18) = F(18) = \frac{1}{175}(18-15)^2 (18-12) \approx \underline{30.8}\%$

In other words, the probability that the project is completed by the contract deadline is 31%.